

(a)

Configuration	MAE ↓	F1-Score ↑	mIoU (%) ↑
w/o Signal Decomposition	0.142	0.823	71.6
w/o Dual-Mask Routing	0.128	0.847	74.2
w/o Angular Guidance	0.135	0.835	73.1
w/o Domain-Balanced Samp.	0.131	0.841	73.8
Full Model (Ours)	0.104	0.892	78.5

(b) Generated material masks

